Data Science Toolbox Portfolio Questions

07 Bayes and Topic Models

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Block 7

Portfolio 07

Portfolio Question 7.1.P - Bayes in Machine Learning

- How is the Bayesian approach used in machine learning?
  - Contrast simple approaches where it can be used to motivate methods, to an area where it is taken seriously.

Portfolio Question 7.2.P - Interpreting LDA

- If we've looked at a collection of documents and a model like LDA has identified some "topics", how would we interpret them?
  - You might start with perplexity and coherence.
  - Do small language models help with interpretability?
  - What about large language models?

Portfolio Question 7.3.P - Applying LDA

- Consider working with a news agency looking to understand the position of its news profile compared to "ABC news" in the workshop, and other news agencies.
  - How could you compare the datasets and why?
  - What would different choices tell you?
  - Be sure to address how evidence-based and interpretable the choices are.